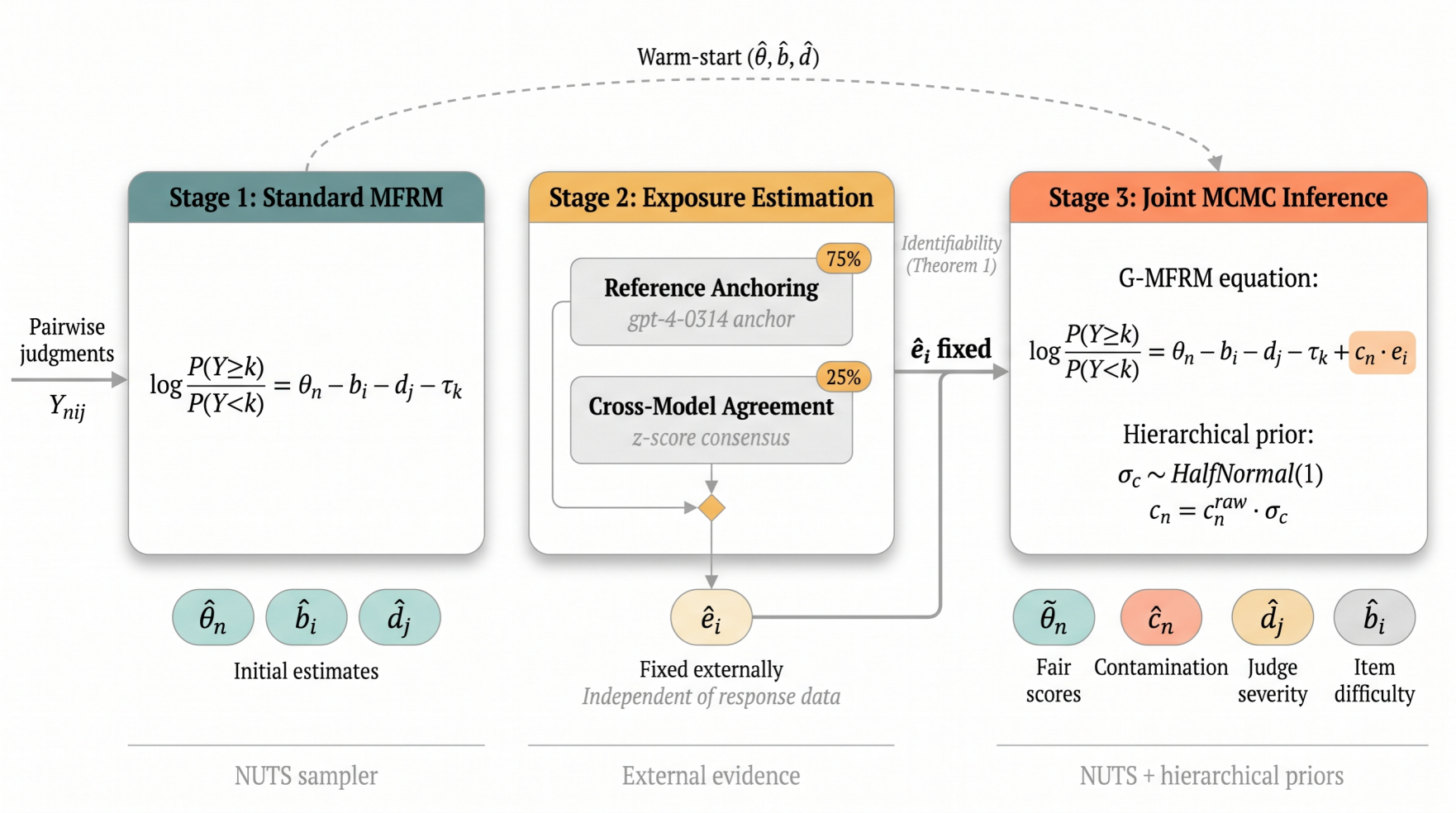

Warm-start $(\hat{\theta}, \hat{b}, \hat{d})$
Stage 1: Standard MFRM
Pairwise judgments
Y_{nij}
$\log \frac{P(Y \geq k)}{P(Y < k)} = \theta_n - b_i - d_j - \tau_k$
$\hat{\theta}_n$ $\hat{b}_i$ $\hat{d}_j$
Initial estimates
Stage 2: Exposure Estimation
75%
Reference Anchoring
gpt-4-0314 anchor
25%
Cross-Model Agreement
z-score consensus
$\hat{e}_i$
Fixed externally
Independent of response data
Identifiability (Theorem 1)
$\hat{e}_i$ fixed
Stage 3: Joint MCMC Inference
G-MFRM equation:
$\log \frac{P(Y \geq k)}{P(Y < k)} = \theta_n - b_i - d_j - \tau_k + c_n \cdot e_i$
Hierarchical prior:
$\sigma_c \sim HalfNormal(1)$
$c_n = c_n^{raw} \cdot \sigma_c$
$\tilde{\theta}_n$ Fair scores
$\hat{c}_n$ Contamination
$\hat{d}_j$ Judge severity
$\hat{b}_i$ Item difficulty
NUTS sampler
External evidence
NUTS + hierarchical priors